

LPWAN sensor networks for continuous environmental compliance monitoring on industrial site rehabilitation projects in Hungary and Belgium.

Public / Government sector × Regulatory compliance and reporting × LPWAN (LoRaWAN, NB-IoT, LTE-M) · OS166 v1 · refreshed 2026-08-18

ATTRACTIVENESS	RIGHT TO WIN	COMPETITION	SERVICEABLE MARKET	HORIZON	PORTFOLIO
60 is the world moving	51 can we play, can we win	MEDIUM 0 named in the evidence	€0 observed evidence · per year	NOW budgeted procurement within 1...	L1 12 signals

Attractiveness and right to win are scored and displayed separately and are never combined (SC-12). Competition is a fourth quantity, not a discount on either.

The opportunity

This opportunity is to provide LPWAN-based environmental monitoring for industrial site rehabilitation projects in Hungary and Belgium, addressing regulatory compliance and reporting needs. Public tenders in these countries signal a concrete demand for such services, and Orange Business can leverage its IoT and digital expertise to deliver a sovereign, certified solution.

Market opportunity

Method	Total (TAM)	Serviceable (SAM)	Obtainable (SOM)	Confidence
Observed: tendered contract value, annualised	€220m €220m – €482m	€0	€0	observed

All figures are annual, in EUR. Ranges compound the uncertainty of each factor below; the base case is the middle column of each.

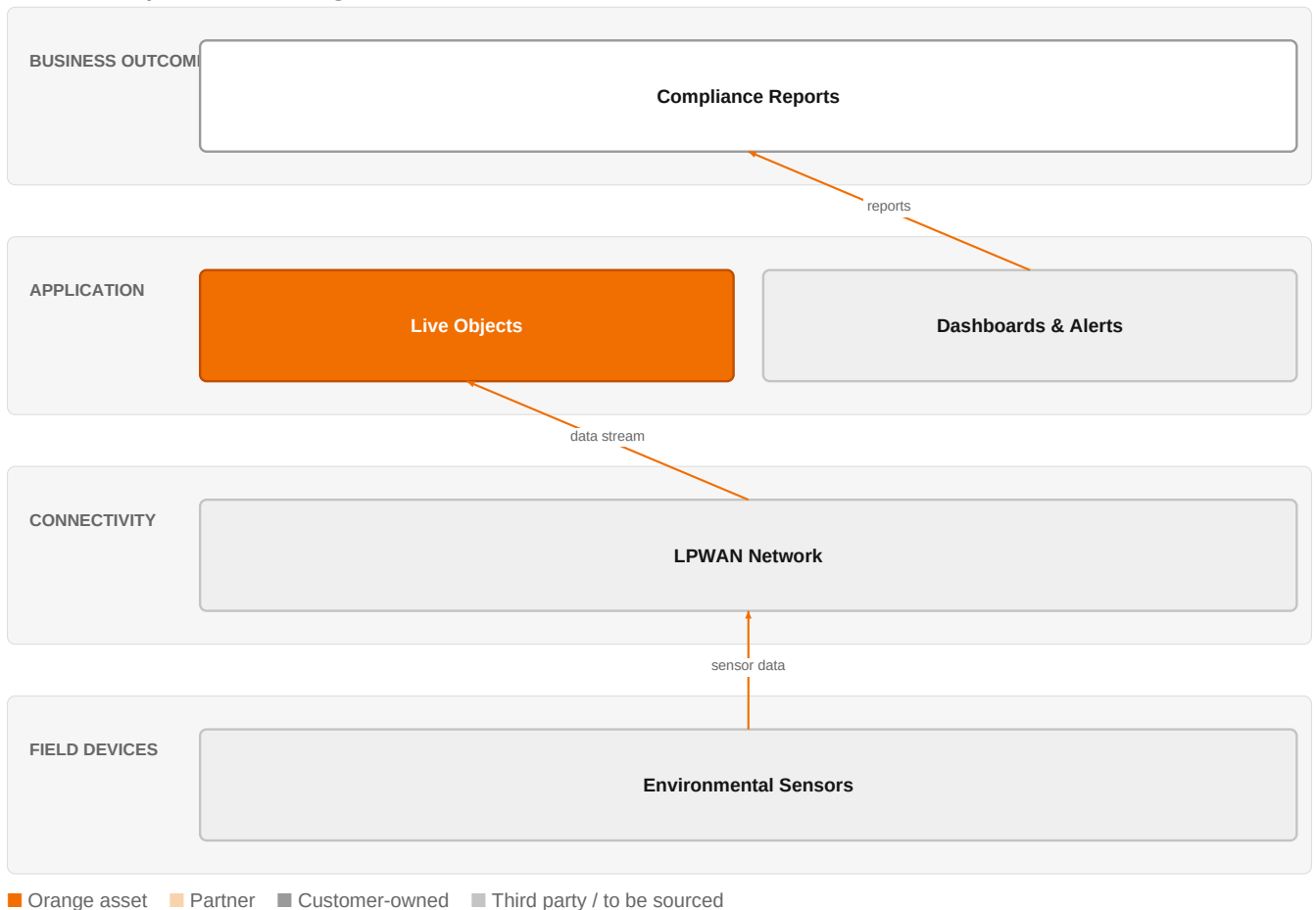
Observed: tendered contract value, annualised

Factor	Value	Basis	Source	As of
Tendered contract value in matching CPV categories	€220m per year	observed	TED (Tenders Electronic Daily) · ted	2026-05-19..2026-08-05
Assumed obtainable share of the serviceable market	1.2 % of SAM	assumption	config/sizing.yaml · obtainable_share	size-2026-08-a

Read this before quoting the number. Observed over 85 days of TED coverage and scaled to a year (x4.3); a short collection window makes that scaling rough. 34 matching notices, matched on vertical via the CPV crosswalk — \$4.5.2's warning applies: a crosswalk error lands directly in this number. Public procurement only. Private-sector demand for the same capability is not visible here at all, so this is a floor rather than a market size.

The solution, in one picture

LPWAN Compliance Monitoring Solution



This diagram shows how sensor data flows from the field through LPWAN connectivity into Orange's Live Objects platform, enabling compliance reporting.

Boxes marked as Orange assets are validated against the linked business graph; a box the model claimed for Orange without a matching asset is shown as third party.

What has changed

Public tenders for industrial site rehabilitation in Hungary and Belgium explicitly include environmental monitoring requirements, indicating a regulatory push for continuous compliance. The tenders reference CPV codes for industrial site rehabilitation and polluted soil treatment, which typically mandate monitoring of air, soil, and water quality. This is part of a broader trend where public authorities are procuring sensor-based monitoring solutions, as seen in similar tenders across Europe for air quality sensors in Germany and vibration sensors for infrastructure in Austria and Italy.

Sources: SIG-8CCC48807679 ted.europa.eu 2026-07-10; SIG-81B837A5B62C ted.europa.eu 2026-06-30; SIG-FF96E1BC7BA5 ted.europa.eu 2026-08-05; SIG-1F577A1F9D8B ted.europa.eu 2026-06-09; SIG-B076E0F3616E ted.europa.eu 2026-05-20

Who buys, and why now

The buyer is typically a public authority or a state-owned remediation company, such as the contracting authorities in the Hungarian and Belgian tenders. The pain is the need to demonstrate regulatory compliance during and after site rehabilitation, which requires continuous, auditable environmental data. The trigger is the publication of a tender for site rehabilitation that includes monitoring obligations, forcing the authority to procure a solution quickly.

Sources: SIG-8CCC48807679 ted.europa.eu 2026-07-10; SIG-81B837A5B62C ted.europa.eu 2026-06-30

Why it is hot now — every claim bound to a dated source

- Public tenders for site rehabilitation in Hungary and Belgium indicate a need for regulatory compliance and environmental monitoring. [SIG-8CCC48807679, SIG-81B837A5B62C]

What Orange would deliver, and why it can win

What Orange would deliver

Orange would assemble a solution using its IoT expertise and Live Objects platform to manage sensor data, with Orange Cyberdefense managed services ensuring security and compliance. The engagement would start with a needs assessment, then deploy LPWAN sensors (via partners) on the site, connect them to Live Objects for data ingestion and processing, and provide dashboards and alerts for compliance reporting. The solution would be hosted on sovereign infrastructure, leveraging Orange's certifications.

Why Orange can win

Orange can win by combining its IoT platform (Live Objects) with its security credentials (ISO 27001, SecNumCloud, CISPE) and its experience in public sector projects (Grand Paris Sud). The ability to offer a fully managed, secure, and sovereign solution differentiates Orange from pure connectivity providers or systems integrators.

Named assets behind this — each individually inspectable (LK-08)

Asset	Type	Link	Why it is linked	Curator
Live Objects	offer	L1	offer provides the technology in a matching domain	unconfirmed
Orange Cyberdefense managed services	offer	L1	offer addresses the use case but not this technology	unconfirmed
ISO 27001	certification	SUP	certification Orange holds is required by this vertical	unconfirmed
SecNumCloud	certification	SUP	certification Orange holds is required by this vertical	unconfirmed
CISPE	certification	SUP	certification Orange holds is required by this vertical	unconfirmed
IoT experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Digital experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Defense and security experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Orange Group researchers	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Grand Paris Sud	reference	SUP	published reference in this vertical, different use case	unconfirmed

Competitive landscape

MEDIUM competitive intensity (score 4.9; 0 of 8 listed players appear in this topic's own evidence)

A real field, including at least one player the customer will already know. Expect to be compared.

Competitor	Category	Position here	Basis	Evidence
Telia Company	Telecom operator peer	sells LPWAN (LoRaWAN, NB-IoT, LTE-M); competes in Connectivity Solutions, OX: Smart Industries	structural	—
Vodafone Business	Telecom operator peer	sells LPWAN (LoRaWAN, NB-IoT, LTE-M); competes in Connectivity Solutions, OX: Smart Industries	structural	—
Actility	Industrial / OT specialist	sells LPWAN (LoRaWAN, NB-IoT, LTE-M); competes in Connectivity Solutions, OX: Smart Industries	structural	—
Accenture	Global systems integrator	active in Public / Government sector; competes in OX: Smart Industries	structural	—
Atos / Eviden	Global systems integrator	active in Public / Government sector	structural	—
Capgemini	Global systems integrator	active in Public / Government sector; competes in OX: Smart Industries	structural	—
Deutsche Telekom / T-Systems	Telecom operator peer	active in Public / Government sector; competes in Connectivity Solutions, OX: Smart Industries	structural	—
IBM	Global systems integrator	active in Public / Government sector	structural	—

evidenced means this topic's own sources name the competitor — the signal ids are in the last column and are dated and clickable in the radar.
structural means the curated register says they sell this technology into this vertical, which is true but is not proof they are in this deal. Register version competitors-2026-08-a.

Qualifying questions for the first meeting

- 1 What specific environmental parameters (e.g., air quality, soil contamination, groundwater) are you required to monitor at the rehabilitation site?
- 2 What is the expected duration of the monitoring period, and how frequently do you need data updates?
- 3 Do you have existing sensor infrastructure or data management systems in place, or is this a greenfield deployment?
- 4 What are the regulatory reporting requirements, and who is the authority that will receive the compliance reports?
- 5 Is there a preference for on-premises vs. cloud-based data hosting, and are there any sovereignty or security certifications required?
- 6 What is the procurement timeline, and are there any budget constraints that would influence the choice between a managed service and a DIY approach?

Objections you will meet

They will say	You can answer
We already have a connectivity provider for our sensors.	That's true, and we can work with your existing provider. However, our value lies in the end-to-end solution: we bring the IoT platform, security, and compliance expertise that ensures your data is reliable and meets regulatory standards.
Your solution seems more expensive than just buying sensors and using a basic platform.	We understand cost is a concern. But consider the total cost of ownership: our managed service includes security, maintenance, and compliance reporting, which reduces your operational burden and risk of non-compliance penalties.
We are not sure about the reliability of LPWAN in rural or industrial areas.	LPWAN is designed for long-range, low-power communication and is well-suited for industrial sites. We have experience with similar deployments and can provide a coverage assessment to ensure reliability.

Risks and unknowns

The main risk is that the customer may already have preferred suppliers or may prioritize lowest cost over value-added services. There is also uncertainty about the specific environmental parameters to monitor and the regulatory framework in each country. Additionally, the timeline for these tenders is unknown, and Orange's ability to partner with local sensor providers is not yet established.

Next action, by role (FR-17)

Role	Do this next
Presales / Proposal	Prepare a bid brief for a public tender on industrial site rehabilitation, anchoring the proposal on Live Objects for sensor data management and Orange Cyberdefense managed services for security, referencing ISO 27001 and SecNumCloud certifications, and using the Grand Paris Sud reference as proof of delivery.
Sales	In a meeting with a public-sector environmental agency in Hungary or Belgium, open with: 'We have helped public entities like Grand Paris Sud manage complex environmental reporting; could we explore how our IoT and cybersecurity capabilities could support your site rehabilitation compliance needs?'
Strategist / Innovator	Commission a deep-dive brief on LPWAN-based environmental compliance monitoring for industrial site rehabilitation, leveraging Orange Group researchers and IoT experts to map the regulatory reporting requirements in Hungary and Belgium.

Evidence

Every claim in this brief resolves to one of these. Tier 1 is an authoritative primary source; tier 4 is vendor-published material, discounted in scoring (§4.3.7).

Signal	Date	Publisher	T	Title and link
SIG-FF96E1BC7BA5	2026-08-05	ted.europa.eu	1	Germany – Sensors – GE grunt, Sensorik, datenqualitatsgesicherter Betrieb, Datenhosting und-Vor...
SIG-3115413B66C4	2026-07-21	ted.europa.eu	1	Finland – Surveillance and security systems and devices – Hoitajakutsu- ja henkilöturvajärjestö...
SIG-D6B20B7CFD20	2026-07-21	ted.europa.eu	1	Czechia – Security cameras – CCTV RLP – integrovaný systém
SIG-8CCC48807679	2026-07-10	ted.europa.eu	1	Hungary – Industrial site rehabilitation – Gyongyosorozsi Erőmű-üzem VII. ütem
SIG-81B837A5B62C	2026-06-30	ted.europa.eu	1	Belgium – Polluted soil treatment or rehabilitation – Rehabilitation du site « Samera » à Sambr...
SIG-B1DA5BAD2988	2026-06-25	ted.europa.eu	1	France – Repair and maintenance services of security equipment – Exploitation et maintenance pr...
SIG-80B4C8BF63DA	2026-06-10	ted.europa.eu	1	Poland – Surveillance and security systems and devices – Dostawa wraz z wymiana i uruchomieniem...
SIG-1F577A1F9D8B	2026-06-09	ted.europa.eu	1	Austria – Sensors – Lieferung von Schwingungssensoren für eine Carboneil-Innovationssektion
SIG-1281187E94DC	2026-05-20	ted.europa.eu	1	Poland – Surveillance and security systems and devices – Dostawa i wdrożenie Systemu klasy XDR ...
SIG-B076E0F3616E	2026-05-20	ted.europa.eu	1	Italy – Sensors – DAC.0129.2025-Fornitura e posa di sistemi di monitoraggio per ponti e viadotti...
SIG-F7CCBA3F83AC	2026-05-19	ted.europa.eu	1	Austria – Sensors – Sensorik zur Passagierstromanalyse
SIG-401D39B133A9	2025-09-16	openalex.org	1	From Counters to Telemetry: A Survey of Programmable Network-Wide Monitoring

How this brief was made

Computed, not written. Attractiveness, right to win, competitive intensity and every figure in the market-opportunity section are computed from stored inputs and can be re-derived (DR-05, NFR-01). No language model produced a number anywhere in this document (§4.4.4).

Written, then checked. The narrative sections and the solution diagram were generated from this topic's evidence and from the named asset and competitor lists. Factual sections must cite signal ids attached to this topic; sections that failed that check, or that contained a generated quantity or an unsupplied organisation, were removed rather than rewritten.

Removed by those checks in this brief: competitive_landscape (no claim survived evidence binding); diagram (1 box(es) claimed to be an Orange asset that was not supplied — shown as third-party instead)

What	Version	When
Scores — weight set	w-2026-08-a	2026-08-18T19:08:26+00:00
Market sizing — assumptions	size-2026-08-a	2026-08-18T21:06:16+00:00
Competitor register	competitors-2026-08-a	2026-08-18T21:06:19+00:00
Narrative — prompt / model	describe-v1 / deepseek-chat	2026-08-19T04:57:40+00:00
Pipeline	0.1.0	2026-08-18

Generated 2026-08-19 04:58 UTC. Scores computed under a different weight set are not comparable with these (SC-10), and neither are market sizes computed under a different sizing version.